

Willhelm Sokolov

Development Daddy, Bootstrap Will

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What's not in your heart is not in your hand...

Education

- 2012–2013 **BS CAE**, *Electrotechnical State University «LETI»*, Saint-Petersburg
- 2013–2015 **BS SE**, *Bonch-Bruевич State University of Telecommunications*, Saint-Petersburg
Image compression (plain C++11), circuitry (Altera's products), distributed-computational system (Qt, math. optimisation)
- 2015–2017 **BS CSE**, *Innopolis University*, Innopolis
DBMS (Qt, Boost), PCG (UE4, C99), UX (Axure PR), technical writing (Markdown, LaTeX)

Bachelor Thesis

- title *GPGPU for High-Performance JSE*
- supervisor Prof. Eugene Zuev (ex. Head of Compilers Lab, Samsung, Moscow)
- description Creation of GPGPU-module for JerryScript (Samsung) using OpenCL/VulkanAPI

Experience

Commercial

- 2025 **Lead Engineer**, *contracts/NDA*, CIS
(9 months) Bootstrapped 2 commercial projects (an educational system, traffic-filtering system) from scratch and audited an in-development payment solution. In all 3 projects, I led a team of 2–5 people (backend, frontend, QA, DevOps).
- Audited an in-development Django+DRF based **payment solution**, overhauled its processes.
 - fixed a six-month old authentication bug in the first week
 - presented a nitpicking technical debt report with priorities and tentative costs
 - conducted thorough code reviews
 - brought development processes and guidelines
 - modernized deployment pipeline: updated to uv and Ruff, added multi-staging to Docker building, cleaned MyPy setup
 - Led an outsource development of an **education system** with government API components (ESIA, Labour Ministry) and emphasis on PII security, including row-level access-control.
 - designed full architecture: C4, ADR, BPMN, UML (ERD, Sequence)
 - communicated directly with: CTO, stakeholders, analytics, tech. team
 - set up GitLab CI/CD: uv, Ruff, MyPy, pytest+coverage, SSH+Docker Compose
 - stack: Django+Ninja, PostgreSQL, RabbitMQ+Celery (background tasks), JWT + ESIA, SSE (notifications), WSS (chat)
 - Led MVP-stage for HTTP/TCP + JS **traffic filtering system**, investor-ready.
 - designed and brought to life a high-performance data pipeline (Nginx+Caddy, pure JS, Taran-tool+Cartridge, Clickhouse, Vector, RedPanda) with real-time monitoring (Grafana+Loki) — RPS were limited by the target service's infrastructure
 - built a monorepo GitHub CD with nested Docker Compose, a single endpoint for configurable deployment, and comprehensive extensive tech. docs
- 2023–2024 **Senior Software Developer**, *NDA*, Worldwide
(1 year) 2M+ users platform development. Worked with multi-platform client and mobile developers, DevOps, and stakeholders.
- design and continuous support for multi-service platform
 - stack: Django/DRF/Jazzmin, FastAPI, PostgreSQL, Docker Compose (v2), Unicorn with Uvicorn workers
 - code quality and environment tools: Rye/PDM, Ruff, MyPy, code reviews
 - feature flags, dispatchers, Django Signals
 - file access race conditioning, IPC, async

- 2022–2023 **Backend Team Lead, Stroyportal, Russia**
(7 months) Moved legacy (6+ years) parts to a new architecture and tools.
- developed a multi-service system from scratch: architecture, instruments, documentation, development process
 - responsible for production deployment and integration with the legacy (strangler pattern)
 - team leading from time to time (80-to-30 dev-lead ratio, not a typo), tasks decomposition, milestones planning
 - stack: async FastAPI, async SQLAlchemy (ORM + Core) + SQLModel, PostgreSQL, RabbitMQ, Dramatiq, pre-commit
- 2021–2022 **Backend Developer, DataArt, USA**
(5 months) Outsource development and internal projects.
- refactoring, designing, and implementing back-end features
 - improving and participated in development processes: code guidelines, white papers, retrospectives, documentation approaches, code reviews
 - designing a Python course for beginners
 - stack: async FastAPI, AWS Chalice, Fast API, DynamoDB, pre-commit
- 2020–2021 **Backend/NLP Developer, RoboVoice, Russia**
(ca. 2 years) I was responsible for improving the back-end part, designing and implementing NLP-related modules, and making the infrastructure for the Data Satanists team from scratch.
- working on NLP-platform (Python3, gRPC, Minio, PostgreSQL, Neo4j, Redis, dramatiq, pytest), ML-infrastructure from scratch (MLFlow, Triton, Hydra, Minio, nginx)
 - consulted the development teams' pipeline (static code analyzer, guidelines, technical debt, retrospectives, etc.), reviewed code
 - refactoring, designing, and implementing back-end features (GoLang, gRPC)
- 2017–2020 **Backend/ML Developer, OWN.space, Germany, Russia**
(2+ years) Mostly I was responsible for improving the existing agents, designing and implementing new ones, and, from time to time, improving the existing submodules architecture.
- NLP, ML, CV: spaCy, NLTK, Tensorflow, Keras, BERT, MS NLP-services
 - developed Science-agent, natural language processing and computer vision R&D
 - refactoring, designing and implementing automatization and support tools
 - team work: git, Scrum, apiary, meetings, team managing (interns)
 - misc.: assisted in UX design, wrote automated tests (CI), consulted HoloLens (MS MR) team
- 2017–2017 **UE4 Developer, GD Forge, Russia**
(5 months) «Cycle» game project: survival, sandbox, craft, quests, fights, buildings.
- gameplay modules:
 - inventory + containers + craft, trauma system
 - dialogue system + workflow tools
 - VFX: UE4 shader tool, HSLS
 - UI: reworking, and implementing visuals and logic for gameplay modules
 - SE: architecture creation
 - team work: git, Trello, meetings
- 2012–2012 **Java Developer, Eureka BPO, Russia**
(4 months) Create, modify, and test modules for Alfresco ECM platform.
- tools:
 - Ubuntu workflow, MS Server
 - Java, MS SQL + PostgreSQL
 - team work: Eclipse + SVN + Jira, meetings
 - web: HTML + CSS+ DOM, e-mail emulating/testing
- Pet-Projects**
- 2025–now **Co-Founder, NDA, worldwide**
Development of declarative language with mode-based visual programming, automated concurrency, and native performance for web-UI platform. We use Rust's language design ecosystem. TBA later.
- 2022–2024 **Project Lead, worldwide**
FRAAND (and front-end) is an open source lending items platform with emphasis on ethics (when it comes to PSA) and user orientation (against B2G or B2B). Think of free (as in libre and cost-less) Avito.

2018–2019 **UE4 Indie-Developer**, *Timenudge Studios*, worldwide

Vertex is a complete re-vamp of «Star Wars: Jedi Knight» series on Unreal Engine 4.

There, I worked on gameplay mechanics design (abilities) and implementation (guns), lore essentials, dev ops (Perforce deployment and support), public relations in CIS.

2016–2018 **Lead Programmer, PM**, *Gray Rat*, CIS

«Star Wars: Republic Commando 2» Unreal Engine 4 game project.

GDD, architecture + tech. design specifications, gameplay modules (UE4 C++ API), script (writing and editing), character rigging (Blender), shaders and level design (WorldMachine, VFX via UE4 particle system), SMM (3K fans).

Skill Matrix

Skill matrix How to read it

■ ■ ■ ■ ■	basic knowledge	■ ■ ■ ■ ■	extensive project experience
■ ■ ■ ■ ■	intermediate knowledge with some project experience	■ ■ ■ ■ ■	deepened expert knowledge
		■ ■ ■ ■ ■	expert / specialist

	Level	Skill	Years	Comment
Language:	■ ■ ■ ■ ■	Python3	8	<i>Thy rule is neigh, GIL!</i>
	■ ■ ■ ■ ■	C++	5	<i>It points at me!</i>
	■ ■ ■ ■ ■	Go	2	<i>Clean to Go!</i>
	■ ■ ■ ■ ■	C	2	<i>I came, I malloc'd, I forgot to free.</i>
	■ ■ ■ ■ ■	Lua	1	<i>Metatable magic.</i>
	■ ■ ■ ■ ■	Rust	1	<i>Deliver this Cargo in time!</i>
	■ ■ ■ ■ ■	TypeScript	1	<i>I Promise I won't React!</i>
	■ ■ ■ ■ ■	Java	2	<i>Do I need a class for that?</i>
System Design:	■ ■ ■ ■ ■	databases	6	<i>CAP and ACID, normalization vs. de-normalization, relational (PostgreSQL), column-oriented (Clickhouse), key-value (Mongo), graph (Neo4j), in-memory (Tarantool)</i>
	■ ■ ■ ■ ■	protocols	6	<i>REST, HTTP (/1.1, /2, QUIC, pipelining, multiplexing), gRPC/RPC, WSS, WebRTC, SOAP</i>
	■ ■ ■ ■ ■	approaches	6	<i>KISS, DRY, SOLID, CRUD vs. CQRS</i>
	■ ■ ■ ■ ■	architecture	5	<i>micro-services vs. monolith, DDD+Clean, ECS, functional, API gateway</i>
	■ ■ ■ ■ ■	computations	4	<i>ticks, scheduling, async ≠ multi-threading ≠ multi-processing ≠ parallelism (FPGA, SIMD, HPC)</i>
	■ ■ ■ ■ ■	messaging	3	<i>event bus, message delivery, RabbitMQ, Redis (pub/sub), Kafka, Redpanda</i>
DevSecOps:	■ ■ ■ ■ ■	servers	3	<i>Nginx/Caddy reverse-proxy, TLS, CORS</i>
	■ ■ ■ ■ ■	security	2	<i>SAST, security headers, OWASP</i>
	■ ■ ■ ■ ■	CI/CD	3	<i>deployment strategies, backwards compatibility, code quality, healthchecks, multi-staged builds</i>
	■ ■ ■ ■ ■	tools	3	<i>Ansible, Docker Compose (v2), Terraform, K8S</i>
OS:	■ ■ ■ ■ ■	Linux	9	<i>BTW, I use Arch. . .</i>
	■ ■ ■ ■ ■	Windows	10	<i>Did you know you can cut telemetrics out?</i>
Team Leading:	■ ■ ■ ■ ■	Communications	6	<i>They will use to love me calling. Or else.</i>
	■ ■ ■ ■ ■	Technical Writing	6	<i>C4, UML, IcePanel, Mermaid.js, Markdown</i>
HCI:	■ ■ ■ ■ ■	UX	2	<i>personas, user study, Axure PR, Figma</i>
	■ ■ ■ ■ ■	Theory	2	<i>Ground theory, human-centred design</i>
Game Dev:	■ ■ ■ ■ ■	Unreal Engine	6	<i>C++ SDK, BP, VFX, IK animation, ECS</i>
	■ ■ ■ ■ ■	Design	3	<i>GDD, vision, TFPS, RPG</i>

Languages

Russian	■■■■■	Native	<i>prose, poetry, essays, technical writing</i>
English	■■■■■	Advanced	<i>fiction and non-fiction writing, oral presentations</i>
Spanish	■□□□□	Elementary	<i>very basics, intuitional reading and listening</i>
Chinese	■□□□□	Elementary	<i>very basics, reading with a dictionary</i>
German	■□□□□	Elementary	<i>very basics, intuitional reading</i>
Attic	□□□□□	Pre-Elementary	<i>basic concepts</i>

Interests

PC Games	Playing, Development, Testing, 12 years
Philosophy	Moism, Buddhism, Antiquity, 4 years
Military Sci.	Ancient Greece, Ancient China, WWII, Combined Arms, modern, 4 years
Dancing	Ballroom, 6 years
Music	Playing (guitar, piano), composing (FL Studio), 2+ years
Singing	Estrade, rap, 3 years
Martial Arts	Muai-Thai, Sambo, Boxing, ACT/Silat-Kali, applied IPSC, 5 years
Hiking	50 clicks a day with 10 kilos, 3K meters high

Articles

- 2014 «Criticism of Voluntary Movements», *XVI International Baltic Communication Forum: GLOBAL AND REGIONAL COMMUNICATIONS: PRESENT AND FUTURE VII*: 59

References

OWN.space

- Sebastian Deneff, PhD HCI
@denef
- Andrey Ermak
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Timenudge Studios

- Chris Papageru, lead
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Raft

- Evgeniy Kokuykin, CTO
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- Alina Lozovskaya, MA CL
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